

Drawn Together: A Collocated Mixed Reality Sketching and Annotation Experience

Andrew Rukangu* Ethan Bowmar† Sun Joo Ahn‡ Beshoy Morkos§ Kyle Johnsen¶

University of Georgia

ABSTRACT

We present Drawn Together, a 3D immersive mixed-reality sketching and annotation tool designed for collaboration research. The system supports a variety of 3D content creation ranging from mid-air, free-form line drawing to voxel-based drawings and mannequins that can be flexibly posed. Groups of users can engage in synchronous, collocated design, education, art, and play aligned against a pass-through video backdrop to allow for physical references and annotations. A number of input options are also presented, including standard motion controllers, a tracked stylus, and ring-augmented hand tracking.

Index Terms: mixed reality, collaboration, annotation, sketching

1 INTRODUCTION

Synchronous, collocated, collaborative mixed reality, where multiple users in the same real-world space can interact with shared 3D virtual elements that blend with the real world, has been a topic of much research for the past several decades [1, 6, 4, 3]. While early works were largely confined to research laboratories, portable systems, including tablets, smartphones, and head-mounted displays, now have sufficient computational hardware and sensors to enable diverse mixed-reality experiences almost anywhere. At the same time, the relative affordability and robustness of modern equipment removes another key barrier for collaborative mixed reality – having multiple devices in the same place simultaneously. Our research aims to exploit these advancements to enable and study in-the-wild collaborative mixed reality experiences.

Drawn Together, designed around the capabilities of the Meta Quest 3 headset, allows small groups of collocated users to engage in collocated, collaborative sketching and annotation activities in mixed reality. While several similar commercial systems exist, creating Drawn Together has enabled us to explore novel ideas for mixed reality collaboration. As a design tool, it fosters creativity and teamwork by allowing virtual objects to be flexibly incorporated into the real-world space. As a research tool, it provides a unique lens into how mixed reality spaces are used for communication and their impacts on team building, learning, and productivity.

2 SYSTEM ARCHITECTURE

Drawn Together was developed using the Unity 3D game engine as a standalone application for Meta Quest 3. While it also works with prior versions, the Quest 3's high-quality stereoscopic color pass-through video helps support the mixed reality effect, and its depth camera enables dynamic occlusion of virtual objects. This feature

allows virtual objects and annotations to naturally appear behind physical elements in the user's environment, enhancing depth perception and fostering the integration of virtual and real-world objects. Such capabilities are increasingly valuable as more users are present in the environment, increasing the likelihood of occlusions.

To maintain alignment with the real world, the system uses spatial anchors to keep two virtual points fixed in the physical space, as otherwise, self-referenced tracking is prone to significant drift. Users can place these anchors in any location, though choosing a real-world landmark, such as marks on the ground, can be helpful. The two landmarks, alongside the world "up" vector that is maintained by the Quest tracking system, are sufficient to establish a world coordinate system. As new users join the session, they can follow the same process, yielding a shared coordinate system[5]. Collaborative mixed reality simplifies this process, as users can point out the landmarks to each other. While it is also possible to automate this alignment on the Quest 3, it requires using the Meta Platform API, which requires internet access. By doing manual alignment, we enable offline deployment and allow for the potential of other headsets and devices that cannot share anchors to be aligned as well.

Network communication is achieved using the open source networking server, VelNet¹ that mimics the widely used Photon Engine server. The VelNet server acts as a broadcast relay and coordinator for multiple clients. A public server hosts internet-connected devices, and private servers can be deployed for offline use. A corresponding open-source Unity3D framework² is then used for customizing the synchronization of individual scene elements as required.

Lastly, Drawn Together includes a web dashboard, shown in Figure 1 (a), and a back-end server and database. These are used to facilitate the uploading and sharing of external assets, saving and loading of scenes, and management of connected headsets, including streaming headset views for observers. For 3D assets, we specifically support the standard GL Transmission format (GLTF) in its compressed form (GLB). A user can upload a GLB file, visualize it, and provide metadata, such as a description and a suggested model scale. The web app offers an application programming interface (API) endpoint providing information about the model, including a download link, which is used by the headset application to load the model.

3 DRAWING AND ANNOTATION TOOLS

Drawn Together's toolkit, shown in the menu of Figure 1 (d), is constantly evolving and currently focuses on exploring various types of tools rather than expanding on existing tool capabilities. The core tool of Drawn Together is mid-air free-form 3D drawing. Using either hand or both together, users can perform brush strokes as shown in Figure 1 (c) that follow a virtual stylus tip that appears to extrude an unlit tubular mesh. We found that an unlit mesh worked best, allowing a low-resolution tube structure and providing high visibility under most circumstances. Various colors and

*e-mail: andrew.rukangu@uga.edu

†e-mail: ethan.bowmar@uga.edu

‡e-mail: sjahn@uga.edu

§e-mail: bmorkos@uga.edu

¶e-mail: kjohnsen@uga.edu

¹<https://github.com/velaboratory/VelNetServer>

²<https://github.com/velaboratory/VelNetUnity>

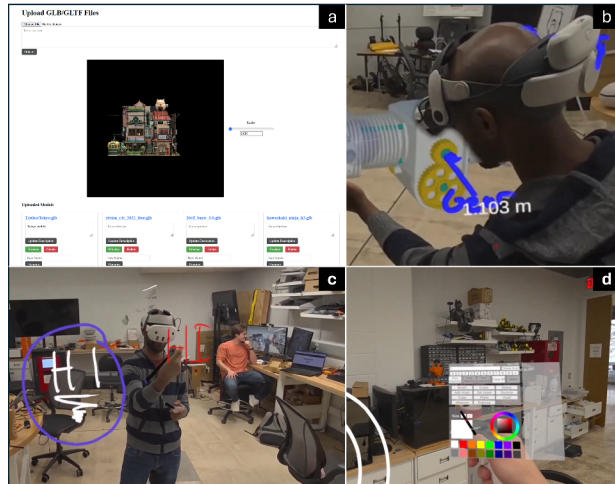


Figure 1: a) GLTF web interface and viewer. b) Object occlusion lends depth to the scene. c) Collaborative annotation and group selection. d) Controller context menu.

a maximum brush size can be selected, with the minimum brush size controllable by an analog input. Strokes are independent 3D objects and can be erased or moved. They (and other objects) can be grouped and moved together or resized.

In addition to strokes, we support simple primitives (e.g., Cubes, Planes), and the import³ of external GLB objects that have been uploaded into the Drawn Together dashboard. Imported objects, as seen in Figure 2 (a), can be elaborate, including many sub-objects and animations. We have also included preliminary support for creating and editing volumetric objects as shown in Figure 2 (b) using an efficient computer-shader-driven approach⁴ within self-contained "cages", a feature we intend upon expanding to support sculpting in addition to drawing. Aside from the above static

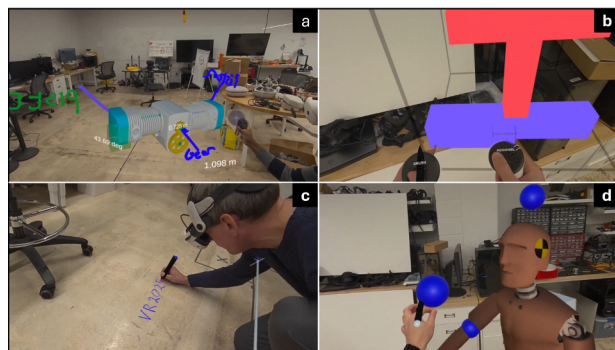


Figure 2: a) Complex problem-solving with 3D models, annotation, and collaboration. b) 3D voxel editing. c) Surface writing stylus support. d) IK Mannequins.

content, Drawn Together features two dynamic tools. The first is a full-body mannequin equipped with an inverse kinematics (IK) system to simplify posing⁵. This mannequin, shown in Figure 2 (d) has been useful for designing human-oriented objects and spaces and as targets for human poses (e.g., dances). More than one mannequin

³<https://github.com/atteneder/gltfFast>

⁴<https://github.com/artnas/UnityVoxelMeshGPU>

⁵Final IK - <http://root-motion.com/>

can be created, and users can puppeteer the mannequin together. The second articulated object is a measuring tool, which consists of three points that show distances and angles, useful for measuring 3D objects (as shown in Figure 2 (a) or real-world objects.

4 INTERACTION DEVICES

The primary tool for Drawn Together is the set of standard tracked Quest 3 touch controllers, which provide numerous digital and analog buttons for quick actions such as resizing, changing tools, bringing up the menu, and selecting objects. However, we have also included support for other controllers, several of which can be used simultaneously. We support hand tracking and using pinch gestures to grab. We also support the recently introduced Logitech MX Ink Stylus, which allows for a more comfortable pen grip and pressure-based writing on physical surfaces, as shown in Figure 2 (c). Finally, we have recently incorporated support for a Bluetooth ring², which increases hand-tracking functionality for drawing and manipulation by providing a physical and capacitive button.

5 DISCUSSION AND CONCLUSION

Thus far, Drawn Together has been used with as many as 12 collocated users, with sessions lasting as long as an hour. System rendering performance is currently the limiting factor, as during demonstrations, users tend to zealously create a large number of strokes and 3D objects. External GLB objects can also be quite complex and reduce performance, which could be addressed with less detail for object generation. However, during more deliberate use with smaller groups, we have found that performance is generally maintained (72 frames per second on Quest 3) throughout the session. Users can also clear their workspace or delete individual objects at any time if they find that performance is suffering. Network performance has not been a limiting factor, especially when used with a local server and dedicated WiFi. For future work, we expect to continue to add new types of tools, especially animation, as well as expand upon the ring-augmented hand-tracking approach. Finally, we aim to support non-collocated collaboration better, as currently, remote users have no context for others' real-world spaces.

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